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Simulation optimization for parcel hub scheduling problem in closed-loop sortation system with shortcuts

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A R T I C L E I N F O

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A B S T R A C T

With the growth of e-commerce, the quantity of delivery orders and parcels is steadily increasing. Reliable and fast parcel delivery services are of utmost importance to customers. To achieve this target, the central parcel consolidation terminals (CPCT) known as parcel sorting hubs are dedicated to improving the operational efficiency especially in the optimal scheduling of inbound trucks. This study aims to determine the optimal schedule and assignment of inbound trucks to the inbound docks, known as the parcel hub scheduling problem with shortcuts (PHSPwS), for minimizing the expected makespan and expected waiting time of inbound trucks in the closed-loop automated sorting conveyor (ASC) system with shortcuts of the parcel sorting hubs. We first construct a mathematical formulation to present the PHSPwS problem based on the traffic control model. However, due to the congestion behavior on the conveyors and stochastic and dynamic nature of the ASC system, the expected objective functions are not analytically available from the mathematical model. Then, a simulation optimization (SO) approach combined with an adaptive genetic-based discrete particle swarm optimization (AGDPSO) algorithm is proposed to tackle the PHSPwS problem in a stochastic and dynamic ASC environment. The closed-loop ASC simulation model is first established to describe the detail operation of the closed-loop ASC sortation system with shortcut. Based on performance evaluation of this simulation model, a novelty algorithm AGDPSO, which hybridized genetic algorithm and particle swarm optimization algorithm with self-tuning parameters is developed to search and find the optimal inbound truck schedule in the large-scale solution space. In particular, the time-varying acceleration coefficients embedded in the AGDPSO are adopted to amplify the effectiveness of the global and local best solutions to obtain a better balance between exploration and exploitation of the searching space. The experimental results reveal that AGDPSO outperforms other algorithms and provides a better and more stable solution. Moreover, the impact of the strategic and operational factors of the closed-loop ASC system for the expected makespan and truck queueing time are also investigated, while the automatic truck unloading mode is confirmed to significantly improve the system efficiency.

1. Introduction

In recent decades, logistics is one of the essential operations in supply chain management that has attracted extensive research

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attention. Logistics comprises planning, implementing, and controlling the movement of goods, services, and related information between the origin point and the specific destinations. As the objective of logistics is to guarantee the delivery of goods on time with high quality and affordable prices, this business process should be managed appropriately. Thus, inbound and outbound logistics run effectively and efficiently. Fig. 1 shows the statistics and forecasts for total ton-miles of freight from 1990 to 2045 in the U.S. It demonstrates that the demand for freight logistics and shipment continuously increased from 2010 to the end of the expected period. To meet the growing needs of new opportunities and business challenges, enterprises strive to improve their delivery potency of material flows and scaling back distribution costs to enhance their market competitiveness in this industry [1]. Thus, it leads the company to have a reliable delivery model.

Generally, two types of delivery networks exist, such as hub-and-spoke and point-to-point. The hub-and-spoke model originated in the transportation industry, where parcels are transported into a network that includes a hub that provides various services and is supplemented by secondary spokes that provide more limited service arrays. Meanwhile, the point-to-point network manages the delivery route from origin to a specific destination without a hub that connects to any other destination. The hub-and-spoke model is commonly used in the parcel delivery industry (PDI) due to it has lower delivery cost than the point-to-point. Fig. 2 illustrates these two delivery networks. Parcels are picked from non-hub terminals and transported to the central parcel consolidation terminals (CPCTs) then brought to another CPCT after parcel sorting. In a CPCT, such as the cross-docking center, parcels from multiple freights are sorted into numerous distinct shipments by destination with almost zero storage at the warehouse [2]. The advantages include increasing the likelihood of full truckloads (FTLs) and shortening the distribution routes. Moreover, it can shorten the order cycle time [3].

An automated sorting conveyor (ASC) has become critical in a CPCT. ASC has many advantages, such as fast operation, can handle a considerable number of items, high reliability, and can handle heavy and massive items [4]. According to the shape of the layout, the ASC layout can be defined into five different types, namely, single row, multi-row, loop layout, open field, and multi-floor [5]. This research concerns the automated closed-loop conveyor system, which is primarily used in the postal industry, and the small land area of CPCT. Fig. 3 explains a closed-loop conveyor system, where the deep gray rectangles represent inbound queues and the white ones represent outbound queues. The loop in the middle shows the material handling systems that connect inbound docks to outbound docks. Some problems may occur in a CPCT, which leads to the lengthy processing time.

Many ways exist to improve the process in CPCT. Several types of research focused on the design of a closed-loop automated sorting system (ASS), which can be called the sortation conveyor layout design problem (SCLDP), such as adding a pre-sorter between the unloading docks and the ASC conveyor and adding shortcuts that act as a new route for the parcel to pass by. The other way is to adopt a scheduling technique, namely, parcel hub scheduling problem (PHSP). This problem is typically found in the PDI. Referring to [6], the objective of PHSP is to come out with a shorter period for sorting the parcels.

Previous studies have made significant contributions to CPCT and ASC. Chen et al. [7] first proposed a method that maximizes the capacity of the ASC by adding shortcuts to closed-loop ASS. Shortcut is similar to a fast lane in the closed-loop ASS, as shown in Fig. 5, it provides alternative routings for the parcel to quickly reach the outbound destination dock without the long transportation of the main conveyor. Chen et al. [7] showed that the construction of a “shortcut” will improve the speed and efficiency of parcel-sorting flows and increase throughput of the ASS system. However, in a closed-loop ASS system with shortcuts, the additional routing determination will increase the complexity of system decision making. Chen et al. [8] further took the ASC with shortcuts design as the background environment and solved the parcel hub scheduling problem by minimizing the sorting makespan. Adding shortcuts not only causes intensifies the complexity of the network, but also results in the congestion at the intersection point of the shortcut and the main conveyor, as shown in Fig. 4. Fig. 4 (b) illustrates the Parcel C in the shortcut must wait until the Parcel A passes before entering the main conveyor. Fig. 4 (c) also exhibits the Parcel C in the shortcut are blocked because Parcels A and B pass through the main conveyor in sequence. The congestion situation may cause the machine capacity of ASC to drop, and the most challenging situation may

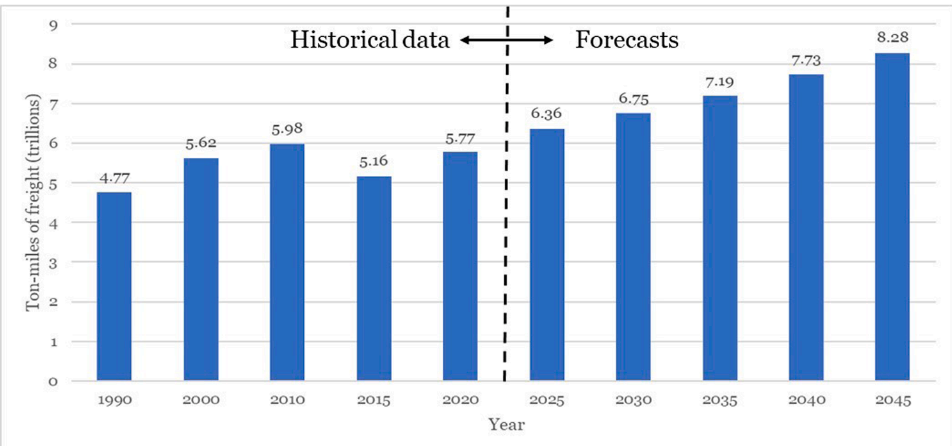


Fig. 1. Historical data and forecasts of total U.S. ton-miles of freight from 1990 to 2045 (trillions) (Bureau of Transportation Statistics & Federal Highway Administration, 2018).

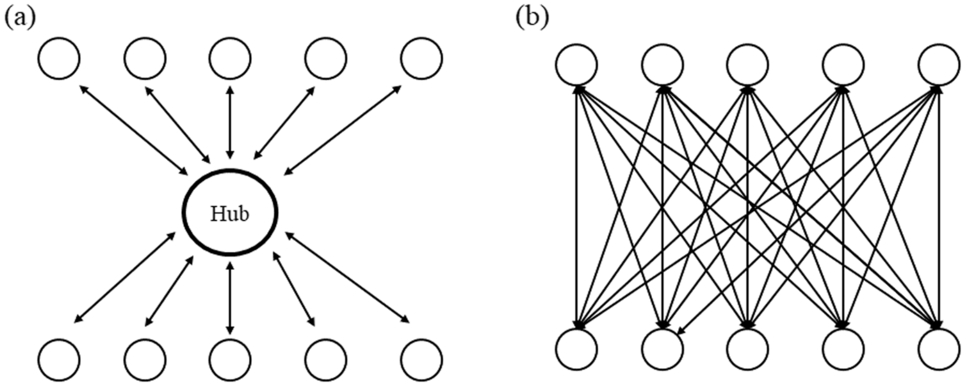


Fig. 2. Illustration of delivery networks. (a) Hub-and-spoke (b) Point-to-point.

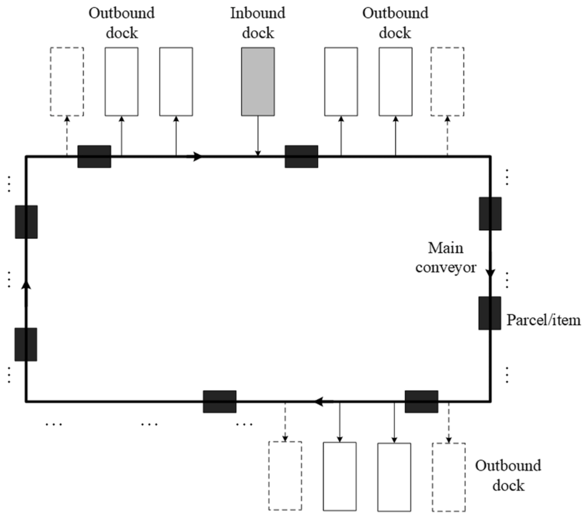


Fig. 3. Illustration of a closed-loop conveyor system.

completely offset all benefits of adding shortcuts. In addition, estimating the severity of congestion using mathematical calculations is difficult. Hence, the system simulation is the best choice, despite its lengthy computing time.

This study aims to determine the optimal schedule of inbound trucks to the inbound docks by minimizing the makespan of the sorting process and queuing/waiting time of inbound trucks in a closed-loop ASC with shortcuts. The makespan of the sorting system is defined as the combination of moving time and idle time. The moving time is the time of transferring parcels from inbound docks to outbound docks through the ASC, whereas the idle time is the delay time due to congestion [6]. A simulation optimization (SO) model based on an adaptive genetic-based discrete particle swarm optimization algorithm (AGDPSO) is proposed to estimate the congestion duration. AGDPSO is used to determine the optimal schedule and some further analysis that would be performed to verify the proposed SO model.

The remainder of this paper is organized as follows. Section 2 presents some previous related works in the fields of PHSP and SO. This section also reveals studies related to PHSP and simulation-optimization algorithms are described in this section. Sections 3 and 4,

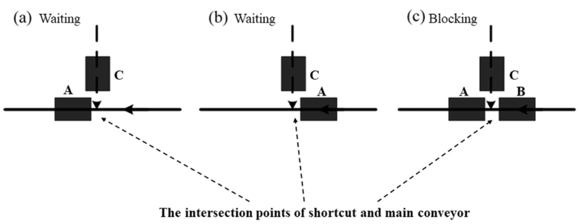


Fig. 4. Illustration of the congestion situation on the ASC.

provide statements, assumptions, and a mathematical model of the parcel hub scheduling problem with shortcuts. Next, Section 5 develops a SO model on the basis of the AGDPSO algorithm to derive the optimal schedule of inbound trucks. Section 6 presents a computational study, which comprises the experimental design and its performance, as well as the representation of the DPSO and simulation model. Section 7 provides conclusions and future research directions.

2. Previous Related Work

PHSP is a common problem in logistics that comprises scheduling a set of inbound trailers to a set of unloading docks. The majority of previous research aimed to reduce the duration of its transfer operations, which include unloading two components, such as inbound trailers and sorting, and the loading process of parcels into outbound trailers. McWilliams et al. [9] developed the PHSP with the goal of maximizing throughput rate by scheduling trailers to unloading docks. As a result, it pushed for the shortest possible time between transfer operations. They focused on freight consolidation terminals in PDI and assumed that only one package per inbound trailer exists. A simulation-based scheduling algorithm (SBSA) that uses a genetic algorithm (GA) was introduced in this research. The output presented that the proposed algorithm generates unload schedules that result in improving operation performance over current practice for freight consolidation terminals in the PDI. Thus, their study contributed to solving practical problems by applying the algorithms.

Many works were conducted in the following years. McWilliams et al. [10] discussed a similar problem. However, they addressed the scheduling of inbound trailers to unload docks at CPCTs in the PDI and considered that the inbound trailers had unequal batch sizes [10]. To solve the PHSP, McWilliams et al. [10] proposed a simulation-based scheduling approach with an embedded genetic algorithm. The findings showed that the proposed scheduling approach was able to reduce the time required to unload the inbound trailers by approximately 3.5% compared to the previous algorithm used and offers solutions that are from 14.7% to 35.8% better than the random approach. However, the optimization via simulation was computationally expensive, and the proposed scheduling approach requires excessive computing times to solve large-scale problems.

McWilliams et al. [11] proposed a mathematical model to solve small-sized problems and a GA for solving large-sized problems. They focused on deriving a workload-balancing solution approach for CPCTs to generate optimally or near-optimal unload schedules while requiring minimum computational time [11]. They concluded that the genetic-based scheduling algorithm (GBSA) offered solutions that were superior to the solutions from SBSA and the random scheduling algorithm (RSA). Although it performed better, the main weakness of the GBSA is that the solution is formulated as a minimax problem, which tends to have numerous non-unique solutions that result in a larger solution space. Furthermore, McWilliams et al. [6] considered the PHSP with unequal batch sizes inbound trailers, which was a combinatorial optimization problem that commonly occurs in a parcel consolidation terminal. In [6], A GA was used to solve the problem of processing many inbound trailers at a significantly smaller number of unloading docks. The experimental analysis unveiled that the algorithm was able to produce solution results that were within 17% of the lower bound, 16% better than a competing heuristic, and 24% better than RSA. The problems were solved by McWilliams et al. [11] and McWilliams et al. [6] by using GAs to obtain minimax workload and the parcel batch sizes on the trailer.

The following research relates to PHSP with various optimization approaches. Chen and Lee [12] captured that the scheduling of inbound product download-and-unpack operations will affect the efficiency of the cross-docking system, as well as that of the total supply chain system. Chen and Lee [12] formulated the problems as a two-machine flow shop problem, where an operation on the first machine was to download and unpack the inbound products while collecting those products with the same destination into an outbound trailer was considered an operation on the second machine. It means that the job at the second machine can only begin after the job at the first machine is completed. The objective was to minimize the makespan. Chen and Lee [12] presented an NP-hard problem, and a polynomial approximation algorithm with an error-bound analysis was also developed. In addition, a branch-and-bound (BB) algorithm was constructed to optimally solve the problems with up to 60 jobs within reasonable times. Chen and Song [13] conducted a study of the two-stage hybrid cross-docking scheduling problem. They considered similar consideration as stated in [12] that the job in the second stage cannot be processed until its precedent subset jobs in the first stage had been complete. The objective was to minimize the makespan by applying mixed-integer programming and CPLEX for small-scale instances. Then, four heuristics methods were proposed to investigate the performance for moderate and large-scaled instances, and the computational experiments were designed to illustrate and compare these approaches.

Boysen et al. [14], Boysen [15], and Briskorn et al. [16] applied heuristic approaches. Considering the problems at cross-docking terminals and realizing economies in transportation costs, Boysen et al. [14] introduced a base model for scheduling trucks at cross-docking terminals, which relies on a set of simplifying assumptions and fundamental insights into the underlying problem's structure, such as complexity. In addition, Boysen et al. [14] developed a building block solution procedure, which can be employed to solve more complex truck scheduling problems. Furthermore, Boysen [15], and Briskorn et al. [16] aimed to minimize the total makespan. Boysen [15] treated a special truck scheduling problem arising in the zero-inventory cross docks of the food industry, where strict cooling requirements forbid intermediate storage inside the terminals. As a result, all products must be instantly loaded onto refrigerated outbound trucks. In [15], the problem was formulated such that different operational objectives, such as the flow time; processing time; and tardiness of outbound trucks, were minimized. Briskorn et al. [16] constructed a short-term deterministic scheduling problem where a finite set of shipments must be loaded onto trays such that the makespan was minimized. In the of model of [16], the different levels of flexibility on how to arrange shipments on the feeding conveyors and distinguish between unidirectional and bidirectional systems were considered. A comparison of the optimization procedures was generated, and performance gains promised by different sorter layouts were also being evaluated.

Other studies correspond to PHSP that applied different heuristics approaches. McWilliams and McBride [17] proposed a beam search scheduling heuristic (BSSH) to solve the PHSP. They aimed to minimize the period of the transfer operation at the transshipment terminal. The BSSH was compared with other scheduling approaches, such as RSA, GBSA, and SBSA. The findings illustrated that for larger size problems, the BSSH was able to produce solutions that were from 4% to 8% of the known optimum solutions. Meanwhile, GBSA and SBSA respectively offered solutions from 23% to 38%, and 6% to 47% of the known optimum solutions. As mentioned in [17], even though GBSA and SBSA offer solutions that are superior to BSSH for small-sized problems, BSSH outperforms these algorithms on large-sized problems requiring much less computational time. Boysen et al. [18] focused on inbound trucks that were unloaded at gates, shipments were identified, sorted by the central sortation conveyor system, and loaded into outbound trailers. Then, it was transported to the next destinations. Boysen et al. [18] attempted to minimize the cost of loading parcels by using the BB technique. In addition, Boysen et al. [18] formalized the resulting optimization problem and provided suited solution procedures. The impact of truck scheduling on the sortation performance of the central conveyor system was tested. Furthermore, Yu and Egbelu [19] proposed a heuristic algorithm (HA) to determine the best truck docking or scheduling sequence for inbound and outbound trucks. The objective function was to minimize the total operations time when a temporary storage buffer to hold items is located at the shipping dock [19].

Other studies were conducted by [20,21], and [22]. Larbi et al. [20] discussed the transshipment scheduling problem in a single receiving and a single shipping door cross-dock under three scheduling policies. First, it was assumed to have complete information on the order of arrivals and the contents of all inbound trucks. The second and third policies assumed the availability of partial and no information on the sequence of upcoming trucks. The model of [20] aimed to minimize the total costs by considering the constraint if part of or all the incoming information is unknown. Next, Bodnar et al. [21] determined the optimal scheduling of inbound and outbound trucks subject to the time windows at multidoor cross dock, which can minimize the operational costs. A mathematical model was derived, and an adaptive large neighborhood search heuristic was proposed in this study. The costs include the costs of handling loads in temporary storage as well as the cost of lateness caused by processing outbound trucks after their respective due dates. The model was implemented on cross-docking scheduling and design problems of four warehouses of a large retailer and shows that substantially smaller cross docks can be used. Heidari [22] applied two meta-heuristics tools, namely, multiple objective differential evolution (MODE) algorithms and non-dominated sorting genetic algorithm-II (NSGA-II) to solve uncertainty problems in

Table 1
Summary of previous works related to PHSP

Previous works	Objectives	Methods	Unique characteristics
Chen, et al. (2019) [8]	Minimize makespan.	AGA	Adding a shortcut layout design.
Heidari, et al. (2018) [22]	Minimize the range of the total service costs & the average total service cost.	NSGA-II & MODE	Vehicle arrival times are unknown.
Briskorn, et al. (2017) [16]	Minimize makespan.	DP	Presorting parcels.
Boysen, et al. (2017) [18]	Minimize the costs of loading parcels to inbound docks.	BS	Different unloading strategies.
Bodnar, et al. (2017) [21]	Minimize operations costs.	HA	Four warehouses of a large retailer show substantially smaller cross-docks.
Mohtashami (2015) [24]	Minimize makespan.	GA	Temporary storage was placed, and inbound vehicles were allowed to enter and leave the dock.
Boysen, et al. (2013) [23]	Minimizing total lost profit	HA, SA	A dock door and a start time had to be assigned to each inbound truck.
McWilliam, et al. (2012) [17]	Minimize the timespan.	HA	A set of inbound trailers each containing a set of heterogeneous parcels to a set of unloading docks.
Larbi, et al. (2011) [20]	Minimize the total cost.	HA	Partial or all the incoming information is unknown.
Boysen, et al. (2010) [2]	Minimize makespan, the processing time of unloading, and tardiness of outbound trucks.	DP	Minimize processing time of unloading.
Boysen (2010) [15]	Minimize the total makespan.	HA	Strict cooling requirements forbid intermediate storage inside the terminals.
McWilliams (2010) [6]	Minimize maximum parcel workload.	GA	Unequal batch size of inbound trailers.
McWilliams (2009) [11]	Minimize the timespan of the transfer operation.	GA	A batch of heterogeneous parcels to a set of unloading docks.
Chen, et al. (2009) [13]	Minimize makespan.	HA, BB	Consider a two-machine flow shop problem.
Chen, et al. (2009) [12]	Minimize the makespan.	NP-hard	The job at the second machine can only proceed after the completion of the jobs at the first machine.
McWilliams, et al. (2008) [10]	Maximize throughput rate.	GA	Equal batch size of inbound trailers.
Yu, et al. (2008) [19]	Minimize operations costs.	HA	A temporary storage buffer to hold items is located at the shipping dock.
Boysen, et al. (2008) [14]	Minimize makespan.	HA	Scheduling relied on a set of simplifying assumptions to derive fundamental insights into the underlying problem's structure.
McWilliams, et al. (2005) [9]	Maximize throughput rate.	GA	The first one was to define the parcel hub scheduling problem (PHSP).

cross docks that were caused by unknown arrival times of the trucks.

Moreover, Boysen et al. [23] proposed HA for minimizing the total lost profit due to unsatisfied trucks by considering an operational truck scheduling problem where a dock door and a start time had to be assigned to each inbound truck. This study settled its computational complexity and developed a heuristic, simulated annealing (SA), to tackle the problem. The two most recent studies in PHSP were [24] and [8]. Mohtashami [24] presented a novel dynamic GA-based method for scheduling vehicles in cross-docking systems, where the total operation time was minimized. This study assumed that temporary storage was placed at the shipping dock and inbound vehicles were allowed to repeatedly enter and leave the dock to unload their products [24]. In this study, some algorithms were proposed, including initialization, operational time calculation, crossover, and mutation for inbound and outbound trucks, independently. Then, a dynamic approach was applied for performing crossover and mutation operations in GA. Chen et al. [8] focused on the parcel hub scheduling problem with shortcuts (PHSPwS), which employed shortcuts to determine the unloading schedule of inbound trailers in the closed-loop sorting process to minimize total makespan. In the proposed AGA, local search (LS) was adopted to avoid entrapment in the local optimal solution, whereas fuzzy logic control (FLC) was performed to adjust the probability of crossover and mutation rates.

Table 1 presents the summary of previous works related to PSHP. Most of previous works focused on the traditional closed-loop conveyor in the sorting process and rarely considered the novel design of ASCs with shortcuts apart from the assignment of delivery groups to the outbound docks. Moreover, sorting parcels may encounter uncertainty factors and unpredictable situations, which have never been considered in the previous literature. To the best of our knowledge, this study is the first that focuses on the parcel hub scheduling problem in a closed-loop sortation system with shortcuts under the stochastic environment with the congestion behavior. In addition, we propose a SO model on the basis of a novelty algorithm, AGDPSO, which hybridized the genetic algorithm and particle swarm optimization algorithm with self-tuning parameters to address this stochastic parcel hub scheduling problem with congestion. This SO model constructs a closed-loop ASC simulation model to show the uncertainty, such as random order in which parcels are unloaded from the truck, accommodates the stochastic human processing time, and shows the severity of shortcuts congestion faithfully. All these constraints were limitedly discussed in the other studies. Furthermore, an AGDPSO optimizer is applied to determine the optimal schedule by considering time-varying acceleration coefficients, amplifying the effectiveness of the global and local solutions to gain a better balance between exploration and exploitation of the searching space.

3. Problem Statement and Assumption

This study focuses on the closed-loop sortation systems with shortcuts, the inbound docks, outbound docks, inbound conveyor and outbound conveyor as shown in Fig. 5. The inbound docks at the distribution center are responsible for unloading parcels from incoming inbound trucks. Then, through the connection of the inbound conveyor, the parcels will be delivered to the main conveyor for sorting. After sorting in the closed-loop ASS, the outbound docks are responsible for unloading the collected parcels to an outbound truck with a specific destination. When the outbound truck is full, it will be delivered to the destination according to the fixed shipping schedule. Therefore, different outbound docks have their corresponding destinations. The outbound conveyor connects between the main conveyor and the outbound dock In addition to the main conveyor, shortcuts provide alternative routes that allow packages to reach outbound docks in a faster and more efficient way. Moreover, the ASC is equipped with shortcuts designed to increase the capacity of the sorting system to accelerate the sorting cycle time. As n trucks containing many parcels with different specific destinations arrived at the distribution center, these parcels must be sorted and placed at the corresponding outbound docks. Because the number of inbound docks is limited, these n inbound trucks are assigned and queuing at m_1 inbound docks by the estimated arrival times, where $m_1 \leq\leq n$. Therefore, How to find the best unloading schedule for inbound trucks in the closed-loop sortation systems with shortcuts is an important topic. This problem is known as the parcel hub scheduling problem with shortcuts (PHSPwS) [8].

This study aims to arrange the order in which the truck enters the sorting center and loads the parcels onto the inbound conveyor belt and complete the sorting processes with the smoothest result. This result is defined as optimal scheduling with the minimum makespan of the sorting process and queuing time in a closed-loop ASC with shortcuts. However, the design of adding shortcuts onto the main conveyor not only increases the capacity of the ASC but also increases the network complexity. Fig. 4 depicts the possibility of

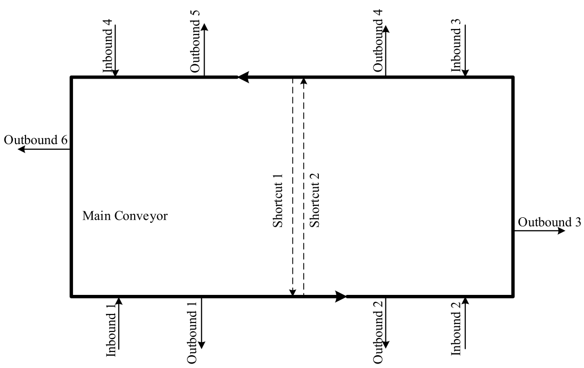


Fig. 5. An Example of the closed-loop ASC system with shortcuts.

a congested situation in which the parcels on the main conveyor are unable to accommodate a new parcel due to a lack of free space. The congestion situation is highly challenging to formulate and predict by a conventional calculation of the analytical model. To address the problem and difficulties, a SO model will be proposed and developed as the solution.

Furthermore, considering the need of verifying the proposed SO model, AGDPSO is applied to determine the optimal schedule and a SO model based on AGDPSO will be proposed. Hence, the following assumptions are implemented in this research:

a Inbound docks are identical, and one inbound truck can only stop at one inbound dock at a time.
b Outbound docks has a specific destination. The operators load parcels from outbound docks to outbound trucks. When the outbound truck is full, it will be delivered to this destination according to the fixed shipping schedule.
c The parcel size and shape is fixed and identical.
d The percentage of all types of parcels to a specific destination is known.
e The unloading sequence of parcels from inbound trucks follows randomness.
f Once the shortcut is full, the parcel will follow the path of the main conveyor.
g The time spent on parcels passing the barcode scanner can be ignored.

In the following section, Table 2 shows the notations, including indexes, parameters, and decision variables that will be used in the mathematical model.

Fig. 5 depicts an example of PHSPwS on the basis of one layer ASC with two shortcuts. The system is equipped with barcode scanners for automatic data capture and identification. Once the trucks arrive, parcels will be unloaded from the vehicles. Then, the parcels are loaded onto the inbound dock. Parcels will pass through the inbound conveyor, where barcode scanners will read the data to determine which layer the parcels will pass through. On the main conveyor, some barcode scanners scan parcels for the second time. The system determines whether the parcels will pass through the shortcuts on the basis of the scanned information. The parcels will be transferred to the corresponding outbound docks once the sorting process is completed. Finally, the parcels will be delivered to the customers.

4. Mathematical Model

This paper refers to the traffic control model of [25] and [26]. In [25], the authors combined the dial-a-ride service constraints with

Table 2
Notation list

Index	
D	Set of trucks, indexed by d
$G(N,A)$	Set of nodes and links
A	Set of links, indexed by $(i,j),(j,k)$
A^{IN}	Set of inbound links, $A^{IN} \subset A$
A^{OUT}	Set of outbound links, $A^{OUT} \subset A$
A^{C}	Set of conveyor links, $A^{C} \subset A$, $A = A^{IN} \cup A^{OUT} \cup A^{C}$
A^{C+}	Set of main conveyor links, $A^{C+} \subset A^{C}$
A^{C-}	Set of shortcut conveyor links, $A^{C-} \subset A^{C}$, $A^{C} = A^{C+} \cup A^{C-}$
N^{R}	Set of inbound dock nodes, indexed by r
N^{S}	Set of destination nodes, indexed by s
T	Set of periods, indexed by t, t'
Parameter	
Q_d^s	Number of parcels on the truck d to the destination s, $s \in S, d \in D$
E_d	Estimated arrival time of truck d to arrive
P	The time required for the truck to park at the dock
F_{ij}	Flow limit of passing into and out the link(i,j), $(i,j) \in A$
L_{ij}	Length of link $(i,j), (i,j) \in A$
V_{ij}	Free speed of the link $(i,j), (i,j) \in A$
$\widetilde{W}_{ij}(t)$	Random congestion speed of link (i,j) at a time $t, (i,j) \in A, t \in T, \widetilde{W}_{ij}(t) \geq V_{ij}$
K	Worst jam density
M	A large number
Decision Variables	
$x_{rd}(t)$	Binary variable, 1 if the truck d is loading at inbound dock node r at time t, otherwise 0; $r \in N^R, d \in D, t \in T$
$e_{rd}(t)$	Binary variable, 1 if truck d starts loading at the inbound dock node r at time t, otherwise 0; $r \in N^R, d \in D, t \in T$
$y_{ijk}^s(t)$	Flow from link (i,j) to link (j,k) and destination to s at time $t, \forall (i,j) \in A \setminus A^{IN}, \forall (j,k) \in A \setminus A^{OUT}, \forall s \in N^S, \forall t \in T$
$z_{ijk}^s(t)$	Binary variable, 1 if there is flow from link (i,j) to link (j,k) and destination to s at time t, otherwise 0; $\forall (i,j) \in A \setminus A^{IN}, \forall (j,k) \in A \setminus A^{OUT}, \forall s \in N^S, \forall t \in T$
p_{rd}	The time needed to load truck d at node r, $r \in N^R, d \in D$
$n_{ij}^{\uparrow s}(t)$	Cumulative count of flow to destination node s at upstream of link(i,j) at time $t, (i,j) \in A, s \in N^S, t \in T$
$n_{ij}^{\downarrow s}(t)$	Cumulative count of flow to destination node s at downstream of link(i,j) at time $t, (i,j) \in A, s \in N^S, t \in T$
cs^s	Completion time at destination s
c	Completion time

the linear program for system optimal dynamic traffic assignment. Hence, it resulted in a congestion-aware formulation of the SAV routing problem. Given the large number of vehicles involved, [25] used a continuous approximation of flow to formulate a linear program. However, another important point of [25] is that the author only considered the interaction between the shared autonomous vehicle (SAV) with travelers at origins and destinations. Thus, they extended their model to consider the flow of parcels on each segment of the whole closed-loop conveyor system from inbound truck arrivals to outbound distributions. The closed-loop conveyor is also divided into networks with multiple links.

As represented in Fig. 5, the closed-loop conveyor network is designed with shortcuts. In this network, links can be divided into four types. Set A^{IN} links inbound docks to the main conveyors, and SetA^{OUT} is the pipeline that sends the parcels out to the shipping dock. The other two are both sorting conveyor belts but are subdivided into the main conveyor belt, A^{C+}, and shortcut conveyor A^{C-}. Also, the main conveyor A^{C+} is assumed to have no traffic jam conditions according to traditional mechanical design. However, the congestion occurs in the shortcut conveyors and set A^{C-}.

This study seeks to simultaneously minimize the maximum completion time and total waiting time of the inbound trucks waiting outside the docks for the sorting process. The following mathematical model is developed to determine the best unloading schedule for the inbound trucks of the parcels.

Objective function

$$Minimum\, E[c] \tag{1}$$

$$Minimum\, E\left[\sum_{d\in D}\sum_{r\in N^R}\left(\sum_{t'\in T} e_{rd}(t') \times (t' - E_d)\right)\right] \tag{2}$$

Subject to

$$\sum_{r\in N^R} e_{rd}(t) = 1 \qquad \forall d \in D, \forall t \in T \tag{3}$$

$$p_{rd} > E_d \times e_{rd}(t) \qquad \forall r \in N^R, \forall d \in D, \forall t \in T \tag{4}$$

$$p_{rd} = \frac{\sum_{s\in N^S} Q_d^s}{\widetilde{w}_{ij}(t)} + P \qquad \forall d, \forall i \in N^R, \forall (i,j) \in A^{IN}, \forall t \in T \tag{5}$$

$$e_{rd}(t) \times \sum_{t'\in[t+1,t+P_{rd}]} e_{rd}(t') = 0 \qquad \forall r \in N^R, \forall d \in D, \forall t \in T \tag{6}$$

$$x_{rd}(t') \times e_{rd}(t) = 1 \qquad \forall r \in N^R, \forall d \in D, \forall t \in T, t \le t' \le t + p_{rd} - 1 \tag{7}$$

$$\sum_{(j,k)\in A\setminus A^{IN}} y_{ijk}^s(t) = \sum_{d\in D} x_{rd}(t) \times \frac{Q_d^s}{p_{rd}} \qquad \forall d \in D, \forall i \in N^R, \forall (i,j) \in A^{IN}, \forall t \in T \setminus \{0\} \tag{8}$$

$$\sum_{(i,j)\in A}\sum_{t\in T} y_{ijk}^s(t) = \sum_d Q_d^s \qquad \forall k = s, s \in N^S \tag{9}$$

$$n_{ij}^{\uparrow s}(t+1) = n_{ij}^{\uparrow s}(t) + \sum_{(i,j)\in A\setminus A^{OUT}} y_{ijk}(t) \qquad \forall (j,k) \in A \setminus A^{IN}, \forall s \in N^S, \forall t \in T \tag{10}$$

$$n_{ij}^{\downarrow s}(t+1) = n_{ij}^{\downarrow s}(t) + \sum_{(j,k)\in A\setminus A^{IN}} y_{ijk}(t) \qquad \forall (i,j) \in A \setminus A^{OUT}, \forall s \in N^S, \forall t \in T \setminus \{0\} \tag{11}$$

$$\sum_{(j,k)\in A^c} y_{ijk}^s(t) \le n_{ij}^{\uparrow s}\left(t - \frac{L_{ij}}{V_{ij}} + 1\right) - n_{ij}^{\downarrow s}(t) \qquad \forall s \in N^S, \forall t \in T \tag{12}$$

$$n_{ij}^{\uparrow s}(0) = 0 \qquad \forall s \in N^S, \forall (i,j) \in A \setminus A^{IN} \tag{13}$$

$$n_{ij}^{\downarrow s}(0) = 0 \qquad \forall s \in N^S, \forall (i,j) \in A \tag{14}$$

$$\sum_{(j,k)\in A^c} y_{ijk}^s(t) \le F_{ij}, \qquad \forall (i,j) \in A \setminus A^{OUT}, \forall t \in T \tag{15}$$

$$\sum_{(i,j)\in A^c}\sum_{s\in N^S} y_{ijk}^s(t) \le F_{jk} \qquad \forall (i,j) \in A \setminus A^{IN}, \forall t \in T \tag{16}$$

$$\sum_{(i,j)\in A^c}\sum_{s\in N^S} y_{ijk}^s(t) \le \sum_{s\in N^S}\left(n_{ij}^{\uparrow s}\left(t - \frac{L_{ij}}{\widetilde{W}_{ij}(t)} + 1\right) - n_{ij}^{\downarrow s}\right) + KL_{ij} \qquad \forall (j,k) \in A^c \setminus A^{C+}, \forall t \in T \tag{17}$$

$$y_{ijk}^s(t) \le M \times z_{ijk}^s(t) \qquad \forall (i,j) \in A \setminus A^{IN}, \forall (j,k) \in A \setminus A^{OUT}, \forall s \in N^S, \forall t \in T \tag{18}$$

$$cs^s \ge t \times z_{ijk}^s(t) \qquad \forall (i,j) \in A \setminus A^{IN}, \forall (j,k) \in A \setminus A^{OUT}, \forall s \in N^S, \forall t \in T \tag{19}$$

$$c \ge cs^s \qquad \forall s \in N^S \tag{20}$$

$$p_{rd} \ge 0 \qquad \forall r \in N^R, \forall d \in D \tag{21}$$

$$n_{ij}^{\uparrow s}(t) \ge 0 \qquad \forall (i,j) \in A, \forall s \in N^S, \forall t \in T \tag{22}$$

$$n_{ij}^{\downarrow s}(t) \ge 0 \qquad \forall (i,j) \in A, \forall s \in N^S, \forall t \in T \tag{23}$$

$$cs^s \ge 0 \qquad \forall s \in N^S \tag{24}$$

$$c \ge 0 \tag{25}$$

$$y_{ijk}^s(t) \ge 0 \qquad \forall (i,j) \in A \setminus A^{IN}, \forall (j,k) \in A \setminus A^{OUT}, \forall s \in N^S, \forall t \in T \tag{26}$$

$$z_{ijk}^s(t) \in \{0,1\} \qquad \forall (i,j) \in A \setminus A^{IN}, \forall (j,k) \in A \setminus A^{OUT}, \forall s \in N^S, \forall t \in T \tag{27}$$

$$x_{rd}(t) \in \{0,1\} \qquad \forall r \in N^R, \forall d \in D \tag{28}$$

$$e_{rd}(t) \in \{0,1\} \qquad \forall r \in N^R, \forall d \in D \tag{29}$$

The first objective (1) aims to minimize the expected makespan $E[c]$ of the system, which equals the time between the start time of sorting the first parcel and the end time of receiving the last parcel. The second objective (2) pursues to minimize the expected total waiting time of the trucks outside the docks. Then, this study adopts the weighted sum method to convert both objectives to a single-objective problem. Section 6.2 shows the detail operations of the weighted sum method. In addition, due the stochastic and dynamic nature of the ASC system, directly calculating the expected value of both objectives through the analytic form is difficult. Therefore, we further develop the SO approach to solve this problem effectively. Equation (3) ensures that all trucks are assigned to one inbound dock, and Equation (4) implies that the loading process will only start after the truck arrives. Equation (5) calculates the processing time required to load parcels from a trailer d onto the inbound conveyor r, which is influenced by the parcel amount on the truck, and the severity of traffic jams. Equations (6) and (7) state that from the starting point of loading the parcels from truck d into inbound conveyor r. Until the end of processing time p_{rd}, the truck d is continuously processed at the conveyor r. This equation is shown in Fig. 6, as an example of a loading schedule from trucks to inbound docks.

Equation (8) exhibits that during the loading process time $p_{rd}$of the trailer d to the inbound conveyor r, all the parcels on d will be sent to the inbound conveyor $(r,j) \in A^{IN}$ on average and stably. Equation (9) ensures that all parcels belonging to the destination s will arrive there. Equations (10) and (11) calculate the cumulative flow quantity through the front-end and tail-end of the link (i,j) at time t. Equations (12) to (14) limit the flow quantity through the front-end and tail-end of the link at time t, which are determined by the link transmission model (LTM).

Flow size into the link (i,j) and passing out the link (j,k) at a time t is limited by flow limit F_{ij} and F_{ik} at Equations (15) and (16). Equation (17) presents the flow size that passes into the link (j,k) at time t during a congestion is limited by the congested seed $\widetilde{W}_{ij}(t)$ and the worst jam density. Fig. 7 illustrates an example of parcel flow at time $(t+1)$, and the link $(i+1, i+3)$ is a shortcut conveyor, 3 parcels pass into, and also 3 parcels pass out the link $(i, i+1)$ at time $(t+1)$; this indicates that $n^{\uparrow s}_{i,i+1}(t+1)$ $n^{\downarrow s}_{i,i+1}(t+1)$ accumulate 3, and $\sum_{s\in S} y^{s}_{i,i+1,i+2}(t+1) = 2$, $\sum_{s\in S} y^{s}_{i,i+1,i+3}(t+1) = 1$. Equation (18), (19) and (20) calculate the makespan of the sorting process when the last parcel passes into the corresponding outbound dock s. Equations (21) to (29) show the domain restriction of all decision variable. Since the proposed mathematical formulation belongs to the stochastic optimization model considering the stochastic and dynamic congestion behaviors (e.g. the truck arrival time, number of parcels per truck, the queue time of the parcels in the ASC system etc.) of the ASC system, it is difficult to directly calculate the expected value of both objectives of the analytic form by the commercial optimization solvers like Gurobi, Cplex, Xpress, etc. To efficiently tackle this complicated stochastic optimization model with mixed binary and continuous variables, we further develop a simulation-optimization approach combining the closed-loop ASC discrete event simulation model with the AGDPSO algorithm. More details regarding the ASC simulation model and the algorithm design are presented in the next sections.

5. Simulation Optimization (SO)

Due the stochastic and dynamic nature of the ASC system, directly calculating the expected value of both objectives through the analytic form shown in Section 4 is difficult. Therefore, we further develop the SO approach to solve this problem effectively. Simulation optimization (SO) is defined as the process of identifying the best input variables from all possibilities without explicitly evaluating each possibility. The output of the simulation model can be the input to the optimization strategy. In addition, the optimization strategy can provide feedback on the process of searching for the best solution. Fig. 8 shows an illustration of the SO concept that was adopted from the framework of [27]. Various algorithms are applied in SO, including Heuristic Algorithm (HA), Meta-heuristics, Response surface methodology (RSM), Gradient-based methods, direct search, and other algorithms.

To determine the best solution, simulation and optimization are applied in this research. The process of proposing a SO is illustrated in Fig. 9. In the optimization phase, the novelty algorithm AGDPSO, which hybridized genetic algorithm and particle swarm optimization algorithm with self-tuning parameters is developed to search and find the optimal inbound truck schedule in the large-scale solution space. Section 5.2 shows the detail operations of the proposed AGDPSO algorithm. In addition, the fitness value of AGDPSO is evaluated by the closed-loop ASC simulation model. In the simulation phase, the closed-loop ASC simulation model is first constructed to describe the detail operation of the closed-loop ASC sortation system with shortcut. On the basis of the solutions found from AGDPSO, the simulation model is executed to evaluate and calculate the single objective value by the weighted sum method. The

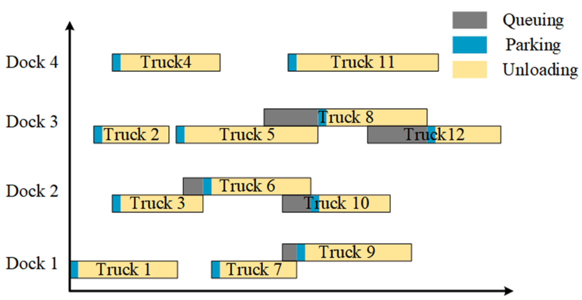


Fig. 6. Example of a loading schedule from trailers to inbound docks.

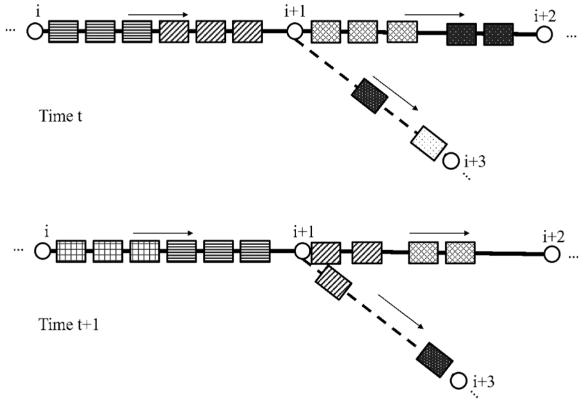


Fig. 7. Example of parcel flow at time *(t+1)*.

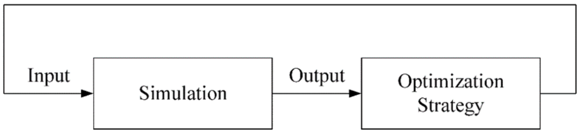


Fig. 8. Concept of simulation optimization [28].

output of the simulation stage will be used to update the global best fitness value among all swarms, Gbest, and local best fitness value of each swarm, Pbest, defined by Section 5.2.2 and search for the best possible solution in AGDPSO. To assess the quality of the schedule and consider the congestion situation that cannot be predicted intuitively, this study builds a model of the simulation sorting system through FlexSim. FlexSim is a 3D dynamic simulation software that is currently used in industries.

5.1. Closed-loop ASC simulation model

A simplified version of a real-world conveyor sortation system is used in this research. The modified layout of the closed-loop ASC sortation system is shown in Fig. 10. The layout includes 8 inbound docks, the main closed-loop conveyor, 4 shortcuts, and 26 outbound conveyors. As a truck arrives at the CPCT, it would wait in front of the determined dock until the dock becomes vacant. Once a truck is parked on the dock, the parcels will be unloaded from the truck and loaded onto the inbound conveyor. Then, the parcels are transported to the closed-loop conveyor and transferred to the corresponding outbound conveyors, which are sorted according to their destination. Given that the main problem of this study is to address the schedule of inbound trucks, the operations after parcels arrive at the outbound conveyor were not considered in the simulation model. Fig. 11 shows the snapshot of the proposed simulation model.

In this work, the existing sortation system is modeled. For representing the real or nearly real to the system, some assumptions and limitations are applied. The first assumption is the time interval of truck's arrival follows the Poisson distribution, which is the most used random distribution in logistics industry. Moreover, the length of the parcel is fixed at 0.5 meters, and the number of parcels per truck follows a normal distribution. The next assumption assumes that all conveyors travel at a fixed speed, and the speed of the main conveyor is 1.25 times faster than inbound conveyors and shortcuts. In a real-world system, sensors on the inbound conveyor scan the zip codes, which determine the destination of the parcels. However, they are ignored in this model.

When a package encounters a fork in the road, that is, to decide whether to take a shortcut, only the shortcut whose endpoint is before the parcel's destination could path through the shortcut. For example, the target of the parcel, which is marked as the yellow rectangle in Fig. 12, is outbound 20. Then, if it fits the pulling range of shortcut 3, so this parcel can take this shortcut. Conversely, if the destination of this parcel is outbound 15, in order not to travel the unnecessary distance, this parcel cannot choose shortcut 3 as the path. It is also worth considering whether if too many packages enter the shortcut at the same time, it will produce worse results. Therefore, different pull rules will be explored at the following section, to discuss whether only part of packages that meet the pulling range enter the shortcut would bring out greater benefits. Next, the optimization algorithm of the proposed SO will be explained in Section 6.

5.2. AGDPSO optimization strategy

Particle swarm optimization (PSO) was proposed by [28]. The preliminary concept of the particle swarm algorithm is a simplified social system, the initial simulation of each generation of correction improvement factors includes nearest neighbor velocity matching, elimination of auxiliary variables, and integration into multidimensional search and distance acceleration [28,29,30]. PSO has various advantages, including fewer parameters for modification, shorter computational time, prevents overlap, and mutates. The main

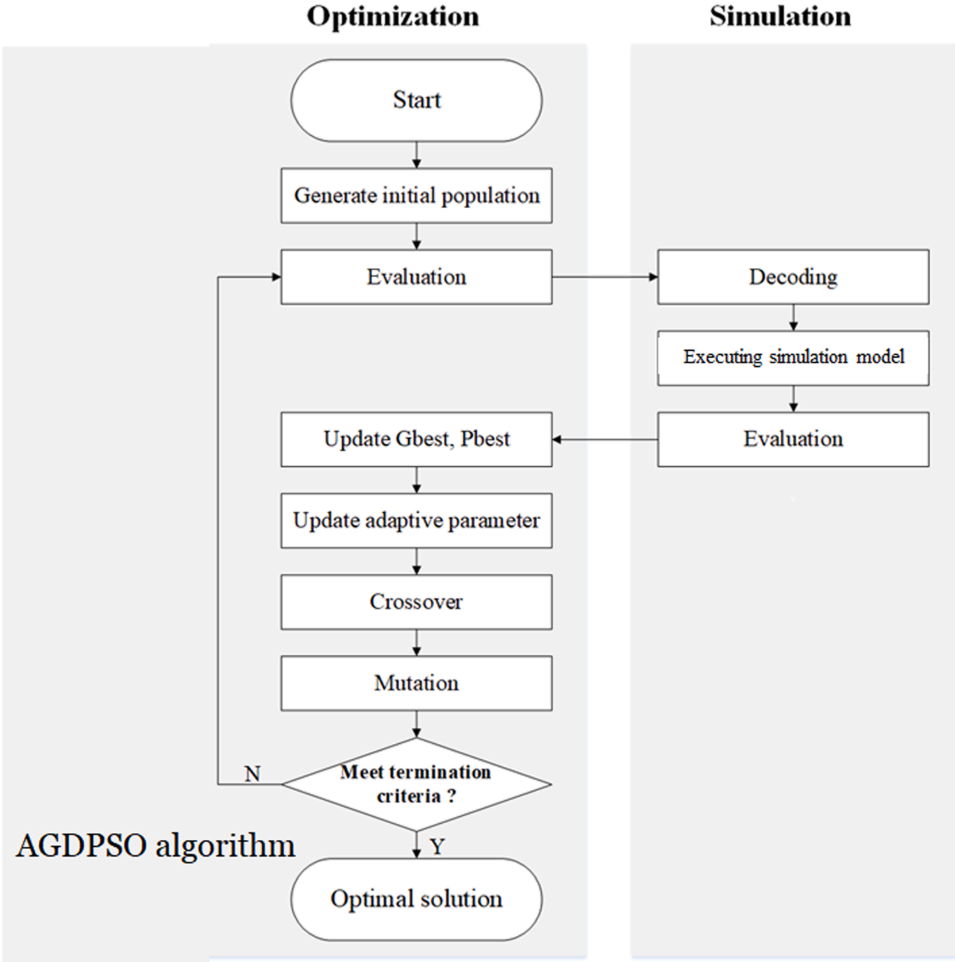


Fig. 9. Proposed SO approach.

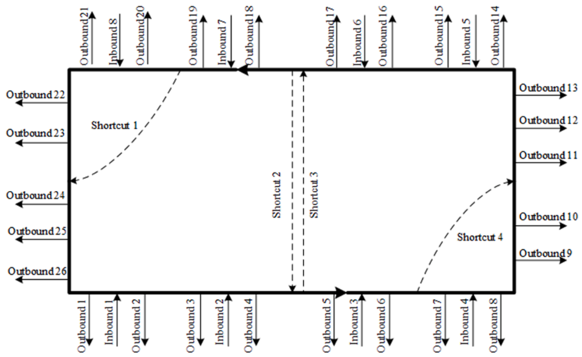


Fig. 10. Proposed closed-loop ASC layout.

concept of PSO is to determine the appropriate flying route and velocity according to the return information of the flock to achieve the best solution.

Previous research of PSO on discrete optimization problems can be easily divided into two categories according to the representation, combinatorial, or binary [31]. In Binary's approach, [32] were widely regarded as the first scholars to apply DPSO to scheduling problems. In addition, the relevant literature included [33,34], and [35]. The most popular method of DPSO is to hybridize GA and PSO, including the use of the two methods sequentially or simultaneously or using GA operators within the PSO framework, such as selection, mutation, and crossover [36]. Lopes and Coelho [37] combined the crossover operator of GA into PSO. Crossover is

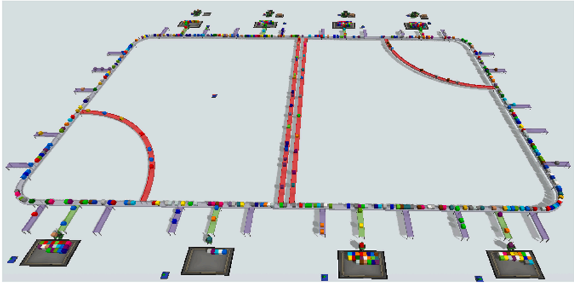


Fig. 11. Snapshot of the proposed closed-loop ASC simulation model.

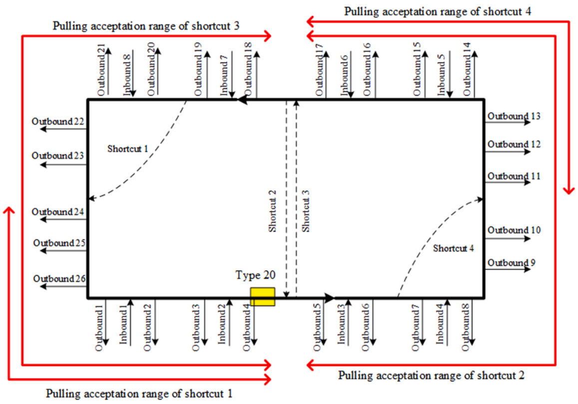


Fig. 12. Illustration of pulling acceptation range for shortcuts.

introduced from the gene algorithm to preserve the presentation of excellent parents to the next generation. In [38], PSO- and GA-based hybrid evolution algorithm (HEA) was developed. Thus, the optimization process time is accelerated by PSO, and the diversity of particles is increased by GA concepts [38].

In this study, AGDPSO is proposed during the optimization phase, setting it apart from similar studies. Table 3 shows the pseudo-code of the proposed AGDPSO. The AGDPSO is the extension of the GDPSO algorithm [39] by considering the self-tuning mechanism of the time-varying acceleration coefficients [40] to amplify the effectiveness of the global and local best solutions to obtain a better balance between exploration and exploitation of the searching space. In addition, instead of using the standard update scheme of PSO, AGDPSO combines current particle, Pbest and Gbest information to apply crossover and mutation operators to update the position of all particles. The following sections present the process of pseudo-code, operation mechanism, and details of this algorithm.

5.2.1. Particle representation and Initialization Population

We adopt the integer-coded solution representation, $x = (x_0, x_1, ..., x_n)$, for the dock assignment of each truck. Then the particles within initial population are randomly generated from the set, N^R, of inbound dock nodes. Due to the longer computational time of SO, this paper sets 30 as the level of population size. Fig. 13 illustrates the one particle of the swarm, when the truck arrives at the

Table 3
Pseudo-code of the proposed AGDPSO

Algorithm 1 AGDPSO

Initialize population P(x) based on the integer-coded representation of the dock assignment of each truck;
While (**! Terminal criteria**) **{**
Evaluation P(x) to calculate their **fitness values** (26)-(30) by the **closed-loop ASC simulation model** constructed in Section 5.1;
Update Gbest, Pbest(x);
Update the parameters c_1 and c_2 by the **time-varying acceleration coefficients** (37) and (38);
For each particle x **{**
Execute **mutation operator**, λ ← G1(x);
Execute **crossover operator** for Pbest(x), δ ← c1⊗G2(λ, Pbest(x));
Execute **crossover operator** for Gbest, μ ← c2⊗G3(δ, Gbest);
Update particle position, x ← μ;
} End for;
} End while;

determined dock. The presentation uses a 1D array whose column represents the truck's index and each random integer from 1 to the number of docks represents which dock the truck will enter. For example, if the 1 column is 3, it means that one truck will arrive before the 3 docks at the scheduled time and will wait until the dock is empty before entering the CPCT and beginnings to unload immediately. Meanwhile, the array length is determined by the number of trucks that will arrive at the CPCT within the duration.

5.2.2. Fitness Value Evaluation and Best value update

After randomly generating the initial population, the fitness value of chromosomes is evaluated. In this research, the fitness value is obtained from the closed-loop ASC simulation model constructed in Section 5.1 in an average of running N times. In addition, the makespan of the system and the average waiting time of the trucks waiting outside the docks are calculated by the weighted sum method (see Eqs. (30)). However, the makespan of the system in Eqs. (31) and the average waiting time of the trucks in Eqs. (32) is the value of two different units or even different scales. Eqs. (33) and (34) show the conversion method of fitness value. By calculating the ratio between the theoretical maximum value and the ideal minimum value, a number between zero and one is obtained. Eqs. (35), (36), and (37) present the weighted parameter that is used to calculate FV. The total of weighted should be 1, and thus each weight should be within the range of 0 to 1.

In this research, the theoretical maximum ofFV_1, which is the upper bound, is the situation when the shortcut is severely blocked. It is not operational and all trucks are assigned to the same dock, resulting in extraordinarily long but extremely unreasonable waiting times. Meanwhile, the best minimum result of FV_2, that is, no blockage, and the sorting process is completed and ended after the last truck arrived at the warehouse.

$$FV = w_1 \times FV_1' + w_2 \times FV_2', \tag{30}$$

$$FV_1 = \frac{1}{N} \times c, \tag{31}$$

$$FV_2 = \frac{1}{N} \times \sum_{r \in N^R} \left(\sum_{t' \in T} e_{rd}(t') \times (t' - E_d) \right), \tag{32}$$

$$FV_1' = \frac{FV_1^{Max} - FV_1}{FV_1^{Max} - FV_1^{Min}}, \tag{33}$$

$$FV_2' = \frac{FV_2^{Max} - FV_2}{FV_2^{Max} - FV_2^{Min}}, \tag{34}$$

$$w_1 + w_2 = 1, \tag{35}$$

$$0 \le w_1 \le 1, \tag{36}$$

$$0 \le w_2 \le 1. \tag{37}$$

Following the evaluation process, the local best fitness value of each swarm, known as Pbest, and the global best fitness value, known as Gbest, among all swarms, will be updated when the new fitness value generated outperforms Pbest and Gbest. Pbest and Gbest are excellent starting points for the swarm's evolution toward the best solution.

5.2.3. Mutation

In this study, we use a swap mutation operator to explore the neighborhood of the current particle position in the AGDPSO algorithm. In addition, it is applied to follow the velocity update rules, in which each coordinate of particles has a certain probability of being mutated. In Equation (38), $G_1()$ is defined as the swap mutation operation. Fig. 14 shows that after the swap mutation, the dock of the fourth truck becomes 6 and the dock of the ninth truck becomes 2.

$$\lambda = G_1(x) \tag{38}$$

5.2.4. Crossover

The PSO-based algorithm is an appropriate method for determining global optimum solution in an optimization problem. Contrastingly, it can be easily trapped into the local optimization when the functions are classified as high dimension. Hence, to tackle the shortcoming in the proposed method, a crossover step is applied to the AGDPSO algorithm to exploit new solution.

In Eqs. (39) and (40), $G_2()$ and $G_3()$ are defined as the crossover operations. First, $G_2()$ combines $Pbest(x)$ and λ obtained from the swap mutation operation to generate the new solution δ using the crossover operation. Then, $G_3()$ combines $Gbest(x)$ and δ to generate

Truck index	0	1	2	3	4		n-3	n-2	n-1	n
Determined dock	1	3	2	3	6	…	1	6	2	4

Fig. 13. An illustration of the integer-coded representation of the dock assignment of each truck.

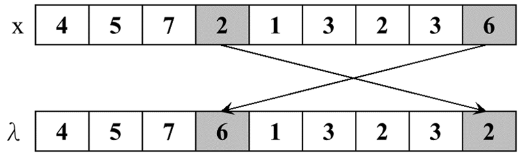


Fig. 14. Illustration of swap mutation operation $G_1()$.

the new solution μ by the crossover operation. Meanwhile,c_1, c_2 represent the crossover rates. A uniform random number r is then generated, and if $r \leq c_1$or $r \leq c_2$, the two individuals undergo recombination. Otherwise, if $r > c_1$or $r > c_2$, the swarm remains unchanged. In this study, the traditional uniform crossover operator is adopted through a binary random array (so-called mask). Fig. 15 shows an example of the uniform crossover. After filtering by the binary mask, The original solution, λ, turns out that the dock assignment of the second car is changed from 5 to 8 in the crossover operation $G_2()$.

$$\delta = c_1 \otimes G_2(\lambda, Pbest(x)); \tag{39}$$

$$\mu = c_2 \otimes G_3(\delta, Gbest); \tag{40}$$

5.2.5. *Time-varying Acceleration Coefficients (TVAC)*

To adaptively regulate the AGDPSO operators, this research utilizes the time-varying acceleration coefficients proposed by [40]. The main idea of the time-varying acceleration coefficients in adaptive AGDPSO is that the parameter will increase or decrease in a determined interval while time passing. Eqs. (41) and (42) present that the increase or decrease of the value of the parameters c_1 and c_2 at each generation g is obtained from subtracting the final value minus the initial value. Then, it is divided by the total generation number G. c_1^s and c_2^s are the initial constant values while c_1^e and c_2^e are the final constant values. In the early stage of the optimization, to accelerate the searching speed, Pbest is a crucial reference value, so the learning curve c_1 is from large to small. However, to converge all swarms at the end, c_2 is expected to be from small to large.

$$c_1 = c_1^e - g \times \frac{c_1^e - c_1^s}{G}, \tag{41}$$

$$c_2 = c_2^s + g \times \frac{c_2^e - c_2^s}{G}. \tag{42}$$

5.2.6. *Terminal Criteria*

For the termination criteria, the maximum number of generations are used in this study. The proposed AGDPSO algorithm will finish predetermined maximum iterations to conclude and stop computing.

6. Computational Study

In this section, a computational study of the proposed model is performed. First, we show the efficiency and effectiveness of the developed AGDPSO algorithm compared to other algorithms including Genetic Algorithm (GA), Greedy Discrete Particle Swarm Algorithm (GrDPSO) that proposed by [41], Adaptive Greedy Discrete Particle Swarm Algorithm (AGrDPSO), Genetic-Based Discrete Particle Swarm Algorithm (GDPSO). Then, through the design of experiments, we investigate the impact of the strategic and operational factors of the closed-loop ASS system for the expected makespan and truck queueing time. The optimal settings of the strategic and operational factors of the ASS system are simultaneously identified.

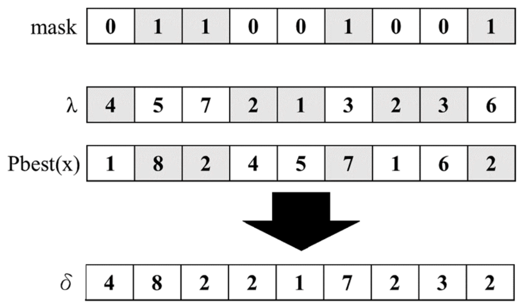


Fig. 15. Example of uniform crossover operation.

6.1. Performance comparison among different optimization strategies

This section explicitly explains the detail of approach, as well as the comparison of five algorithms that are commonly used, such as GA, GrDPSO that proposed by [41], AGrDPSO, GDPSO, and AGDPSO. All of GrDPSO, AGrDPSO, GDPSO and AGDPSO belongs to the extensions of the discrete particle swarm algorithms (DPSOs). The definition of five algorithms are exhibited in Appendix A1. All these algorithms have the same method of generating the initial population, evaluating the fitness value, and encoding the swarms. Moreover, these DPSOs utilize similar methods of the crossover, which are One-Point Crossover. The main difference among these DPSOs is the selection of swarms, mutation method, and the adaptive time-varying acceleration coefficients. In addition, Appendix A2 clarifies the differences in processes and mechanisms between GA, GrDPSO, AGrDPSO, GDPSO, and AGDPSO. Moreover, Table 4 shows the parameter setting of optimization for each algorithm in this research.

In order to further compare the performance of different algorithms, we change the arrival time of different trucks from large value to small value as different instances. According to the comparison results in Table 5 and 6, the AGDPSO algorithm outperforms other four algorithms under all instances on the makespan and truck queuing time. The figures of the performance comparison among different algorithms are shown in Appendix A3. Moreover, in most instances, the DPSO-based algorithms perform better than the GA algorithm. As the arrival-time interval becomes smaller, the makespan and truck queuing time also gradually increase. Therefore, the proposed simulation optimization model based on the AGDPSO algorithm can effectively solve the stochastic parcel hub scheduling problem with shortcuts.

6.2. Design of Experiments (DoE)

In this work, many factors are being considered, which can be classified into situation factors, strategy factors, or ASS design. Situation factors come from the characteristics of the parcel itself, including the arrival-time interval and the distribution balance of parcels. Strategy factors define the configuration policy within CPCT, including the process of unloading trucks and the number of parcels per truck. ASS design mainly presents the speed of the conveyor and the pulling rule of the shortcuts. In this section, we investigate the impact of the strategic and operational factors of the closed-loop ASS system for the expected makespan and truck queueing time through the design of experiments. The details of these factors are explained as follows:

- Situation factors

The arrival-time interval of the truck follows the Poisson distribution. Moreover, the average arrival time will be the first situation factor; the truck will arrive at CPCT at a given average time interval during the duration, the 0 to 1440 minutes. The unloading order of parcels is random. Furthermore, two levels of destination are utilized, which are balanced and unbalanced. For the balanced destination distribution, the average number of parcel on each truck that must be sent to each destination is subject to uniform distribution. Conversely, if the destination of most packages on the truck is destined on some specific destination docks, then it is called

"unbalanced destination." For example, 75% of packages in a truck is destined for docks on the same side of the closed-loop ASS system. This study sets the unbalance destination at 50%–75%.

- Strategy Factors

Many workforces are used in traditional CPCT, but automation is an inevitable trend in modern times. Therefore, this study adopts two truck-unloading methods, workforce, and mechanical arm. Among them, the mechanical arm is relatively fast, but it also has a relatively high costs. Another issue that must be discussed is the average number of parcels in each truck. This factor is affected by the volume of the parcels and the storage capacity of the truck. This study assumes that the number of parcels in each truck obeys the normal distribution.

- ASC Design Factors

This research assumes that the ASC layout design within the CPCT is fixed, as shown in Fig. 10. The speed of the conveyor and the pulling rule of shortcuts are discussed for improving ASC performance. Congestion will occur at the destination of the parcels due to the large number of parcels entering the shortcut simultaneously, which also reduces the shortcut's original effect. Hence, some

Table 4
Parameter settings for each algorithm

Algorithm	GA	Algorithm	GrDPSO	AGrDPSO	GDPSO	AGDPSO
Population	30	Population	30	30	30	30
Generation	100	Generation	100	100	100	100
Crossover rate	0.80	w	0.78	0.78	0.60	0.60
Mutation rate	0.05	c_1 (initial)	1.49	0.72	0.50	0.95
		c_1 (end)	-	0.00	-	0.00
		c_2 (initial)	1.49	0.48	0.50	0.40
		c_2 (end)	-	1.24	-	0.95

Table 5
The makespan of different optimization strategies under various arrival-time intervals

Arrival-time interval (min)	instance	GA	GrDPSO	AGrDPSO	GDPSO	AGDPSO
20	eab1	1,671.90	1,675.73	1,665.13	1,663.97	1,621.63
	2	1,652.97	1,679.50	1,688.52	1,675.87	1,487.57
	3	1,510.53	1,475.37	1,483.57	1,520.97	1,480.43
	Avg.	**1,611.80**	**1,610.20**	**1,612.41**	**1,620.27**	**1,529.88**
10	4	1,835.70	1,663.97	1,822.43	1,744.40	1,652.97
	5	1,770.70	1,652.40	1,713.30	1,698.23	1,647.87
	6	1,713.17	1,625.37	1780.60	1,689.00	1,638.83
	Avg.	**1,773.19**	**1,647.25**	**1,772.11**	**1,710.54**	**1,646.56**
5	7	2,264.03	2883.73	1,758.33	2,051.37	1,751.70
	8	2,390.67	1,738.67	1,702.03	1,952.30	1,696.30
	9	2,158.07	1,716.43	1,728.73	1,915.60	1,721.53
	Avg.	**2,270.92**	**2,112.94**	**1,729.70**	**1,973.09**	**1,723.18**

Unit: minute; minimum value is marked by underline

Table 6
The truck queuing time of different optimization strategies under various arrival-time intervals

Arrival-time interval (min)	instance	GA	GrDPSO	AGrDPSO	GDPSO	AGDPSO
20	1	6.30	1.37	3.15	0.78	1.42
	2	2.56	0.28	2.02	0.41	1.56
	3	3.98	0.82	1.17	0.59	0.99
	Avg.	**4.28**	**0.82**	**2.11**	**0.59**	**1.32**
10	4	28.89	5.03	25.29	7.94	5.05
	5	17.83	3.22	10.74	3.98	2.69
	6	13.66	4.65	15.14	5.08	4.65
	Avg.	**20.13**	**4.30**	**17.06**	**5.67**	**4.13**
5	7	450.37	471.77	26.13	102.81	21.44
	8	528.07	390.61	27.51	126.79	27.61
	9	386.50	11.82	16.71	68.65	17.88
	Avg.	**454.98**	**291.40**	**23.45**	**99.41**	**22.31**

Unit: minute / truck; Minimum value is marked by underline

judgement logic will arise when using the shortcuts for routing:

- Pulling rule A, all parcels with available destinations will pass through the shortcuts.
- Pulling rule B, under the conditions of the destination, the parcel will take the shortcuts under a given probability PLP_B.PLP_B is set as 50% by one-way ANOVA experiment.
- Pulling rule C, under destination conditions, only if the shortcut is not congested when the parcel arrives in front of it.
- Pulling rule D, under the conditions of the destination, only if the density, defined by Equation (43), of the main conveyor is lower than the given value, PLD_D.PLD_D is set as 25% by one-way ANOVA experiments.
- Pulling rule E, under the conditions of the destination, only if the density, defined by Equation (43), of the main conveyor is lower than the given value, PLD_E.PLD_E is set as 75% by one-way ANOVA experiments.

$$density = \frac{(number\ of\ parcels\ on\ the\ conveyor)}{(conveyor\ length \div average\ parcel\ size)} \tag{43}$$

Table 7 shows the summary of the factors and levels of the proposed experiments. The experimental parameters of this study, N is set as 30; that is, the fitness value of each swarm is the average of 30 runs. In addition, in terms of the weight of the two targets, w_1 is 0.6

Table 7
Levels of the factors

Factors	Levels				
Arrival-time interval (min)	5	20			
Balance Destination	Non-balance	Balance			
Unloading method	Human (1.5min/parcel)	Automation (0.5min/parcel)			
Average batch size (parcel/truck)	50	75			
Conveyor speed (m/s)	1.5	2.0			
Pulling rule of shortcut	A	B	C	D	E

and w_2 is 0.4, according to one-way ANOVA results.

6.2.1. ANOVA result of DOE

Table 8 shows the ANOVA table of DoE results for the makespan. The significance level is set at 95% (α = 0.05). Fig. 16 demonstrates the main effect plot for the makespan. As shown in Fig. 18, most factors, including truck unloading mode, average batch size, average arrival-time interval, the balance of destination, and conveyor speed, significantly affect the makespan, except the pulling rule of shortcuts. The impact of the pulling rule may be overly small and is dominated by other factors. Nonetheless, Chen et al. [8] have demonstrated that shortcuts-based ASC layout can greatly improve the sorting efficiency and decrease the makespan compared to shortcut-free layout. In addition, from the experimental results, we can see that the pulling rules A and C mostly perform better than other rules from Tables 9–11. This means that the destination of the package can be delivered through the shortcut as much as possible, which will also reduce total makespan of the ASS system. These two pulling rules can also be recommended as important references for practical ASC systems in designing decision rules for controlling diversion. Among the factors with significant influence, the truck unloading mode has the most significant effects, which implies that enhancement in the unloading process can significantly improve the system's efficiency. Furthermore, automation is the right development direction in the future.

Fig. 17 represents the interaction plots for the makespan. Interaction plots are constructed by plotting both variables together on the same graph, and the distinguishing feature of strong interactions is the degree of non-parallelism between two lines. Fig. 19 indicates that the interactions are mostly significant for truck unloading mode*average batch size, truck unloading mode*average arrival-time interval, truck unloading mode* balance of destination, truck unloading mode*conveyor speed, average batch size* average arrival-time interval, average arrival-time interval* balance of destination, average arrival-time interval *conveyor speed. Among them, a strong interaction exists between truck unloading mode and average arrival-time interval in terms of the makespan. Consequently, the human-based truck unloading mode coupled with the intensive of truck arrivals will seriously result in extremely long makespans.

6.2.2. Optimal setting under different factors

Tables 9–11 demonstrate the optimal setting of strategy factors and ASC design factors under different situation factors (average arrival-time interval and balance destination) for different objective values, including FV, makespan, and average truck queueing time. We discovered that the automation of truck unloading and acceleration of conveyors can reduce makespan and average truck queueing time. In addition, the number of trucks arriving per hour has increased in the opposite direction from the batch size of parcels that can be expected in the truck. Consequently, it is reasonable to assume that the system has reached its maximum capacity. Finally, pulling rules A and C dominate other pulling rules for various objective values.

7. Conclusions

This study investigates the parcel hub scheduling problem with shortcuts considering the congestion behavior on the conveyors and stochastic and dynamic nature of the ASC system. The PHSPwS problem aims to determine the optimal schedule and assignment of inbound trucks to the inbound docks for minimizing the expected makespan and expected waiting time of inbound trucks in the closed-loop ASC system with shortcuts of the parcel sorting hubs. A mathematical formulation based on the traffic control model is first constructed to present the PHSPwS problem However, due to the stochastic and dynamic nature of the ASC system, the expected objective functions are not analytically available from the mathematical model. Then, the SO approach combined with the novelty AGDPSO algorithm is proposed to tackle the PHSPwS problem under the stochastic and dynamic ASC environment. The experimental results demonstrate that AGDPSO outperforms other algorithms including GA, GrDPSO, AGrDPSO and GDPSO and provides a better and more stable solution. Furthermore, through the design of experiments, we investigate the impact of the strategic and operational factors of the closed-loop ASC system for the expected makespan and truck queueing time. The optimal settings of the strategic and operational factors of the ASC system are simultaneously identified. Among them, the automatic truck unloading mode has been proven to significantly improve the system efficiency.

The contribution of this study is to define and solve the parcel hub scheduling problem with shortcuts by the reliable simulation optimization approach. The constraints of the detail operation of real-world ASC systems are also highly considered in the closed-loop ASC simulation model. Given that the closed-loop ASC system is equipped with shortcuts, the congestion phenomenon becomes critical in this study and makes the problems more complex than in previous studies. This study used simulation to show the uncertainty of the random order in which parcels are unloaded from the truck and the severity of shortcut congestion faithfully, which was rarely proposed in the other previous studies. Furthermore, a novelty AGDPSO is developed and improved by considering the self-tuning mechanism of the time-varying acceleration coefficients. It enables parameters to be adjusted with generation, increasing the effectiveness of global and regional best optimal to strike a balance between exploring and exploiting the searching space. In addition, instead of using the standard update scheme of PSO, AGDPSO also combines current particle, Pbest and Gbest information to apply crossover and mutation operators to update the position of all particles effectively.

Concerning the characteristics of various arrival intervals of trucks and unequal destination distribution, this research provides a solution for the PDI facing a rapid growing demand. Due to some limitations, this study still warrants improvement. Further analysis related to this study must be conducted. This study can be extended in the following directions:

- This study assumes that the package will be removed immediately after passing the outbound conveyor. Hence, different outbound characteristics can be considered in the future.

Table 8
Summary of DoE results

Source	F-Value	P-Value
Model	94.94	0.000*
Linear	1288.25	0.000*
Truck unloading mode	3329.60	0.000*
Avg. batch size (parcel/truck)	1066.84	0.000*
Avg. arrival-time interval (min)	7114.85	0.000*
Balance destination	19.08	0.000*
Conveyor Speed	58.78	0.000*
Pulling Rule	1.28	0.276
2-Way Interactions	112.51	0.000*
Truck unloading mode*Avg. batch size (parcel/truck)	172.95	0.000*
Truck unloading mode*Avg. arrival-time interval (min)	2361.25	0.000*
Truck unloading mode*Balance destination	11.52	0.001*
Truck unloading mode*Conveyor Speed	25.09	0.000*
Truck unloading mode*Pulling Rule	1.39	0.237
Avg. batch size (parcel/truck)*Avg. arrival-time interval (min)	748.04	0.000*
Avg. batch size (parcel/truck)*Balance destination	0.53	0.469
Avg. batch size (parcel/truck)*Conveyor Speed	1.62	0.205
Avg. batch size (parcel/truck)*Pulling Rule	0.50	0.738
Avg. arrival-time interval (min)*Balance destination	14.90	0.000*
Avg. arrival-time interval (min)*Conveyor Speed	28.07	0.000*
Avg. arrival-time interval (min)*Pulling Rule	0.77	0.548
Balance destination*Conveyor Speed	0.10	0.755
Balance destination*Pulling Rule	0.10	0.983
Conveyor Speed*Pulling Rule	0.06	0.993
3-Way Interactions	2.18	0.000*
Truck unloading mode*Avg. batch size (parcel/truck)*Avg. arrival-time interval (min)	54.16	0.000*
Truck unloading mode*Avg. batch size (parcel/truck)*Balance destination	3.03	0.083
Truck unloading mode*Avg. batch size (parcel/truck)*Conveyor Speed	5.46	0.020*
Truck unloading mode*Avg. batch size (parcel/truck)*Pulling Rule	0.23	0.924
Truck unloading mode*Avg. arrival-time interval (min)*Balance destination	10.42	0.001*
Truck unloading mode*Avg. arrival-time interval (min)*Conveyor Speed	20.46	0.000*
Truck unloading mode*Avg. arrival-time interval (min)*Pulling Rule	1.06	0.375
Truck unloading mode*Balance destination*Conveyor Speed	1.45	0.230
Truck unloading mode*Balance destination*Pulling Rule	0.26	0.902
Truck unloading mode*Conveyor Speed*Pulling Rule	0.25	0.911
Avg. batch size (parcel/truck)*Avg. arrival-time interval (min)*Balance destination	0.58	0.448
Avg. batch size (parcel/truck)*Avg. arrival-time interval (min)*Conveyor Speed	2.14	0.144
Avg. batch size (parcel/truck)*Avg. arrival-time interval (min)*Pulling Rule	0.53	0.716
Avg. batch size (parcel/truck)*Balance destination*Conveyor Speed	0.13	0.721
Avg. batch size (parcel/truck)*Balance destination*Pulling Rule	0.12	0.975
Avg. batch size (parcel/truck)*Conveyor Speed*Pulling Rule	0.01	1.000
Avg. arrival-time interval (min)*Balance destination*Conveyor Speed	0.03	0.855
Avg. arrival-time interval (min)*Balance destination*Pulling Rule	0.09	0.987
Avg. arrival-time interval (min)*Conveyor Speed*Pulling Rule	0.03	0.998
Balance destination*Conveyor Speed*Pulling Rule	0.17	0.955
4-Way Interactions	0.33	1.000
5-Way Interactions	0.13	1.000
6-Way Interactions	0.05	0.995

*Indicates the control factor has the significant impact on the makespan while its p-value is less than $\alpha = 0.05$

- This study only considers the scheduling of trucks entering the CPCT. In future studies, it may be combined with other CPCT tasks such as human resource allocation, truck scheduling, and CPCT departure decision.
- Improve the proposed AGPSO of this study or combine it with another algorithm to enhance the quality and robustness of the results.
- Develop other algorithms instead of DPSO and compare the performance of each algorithm.

Data availability

No data was used for the research described in the article.

Appendix A

A.1. Detail contents of five algorithms

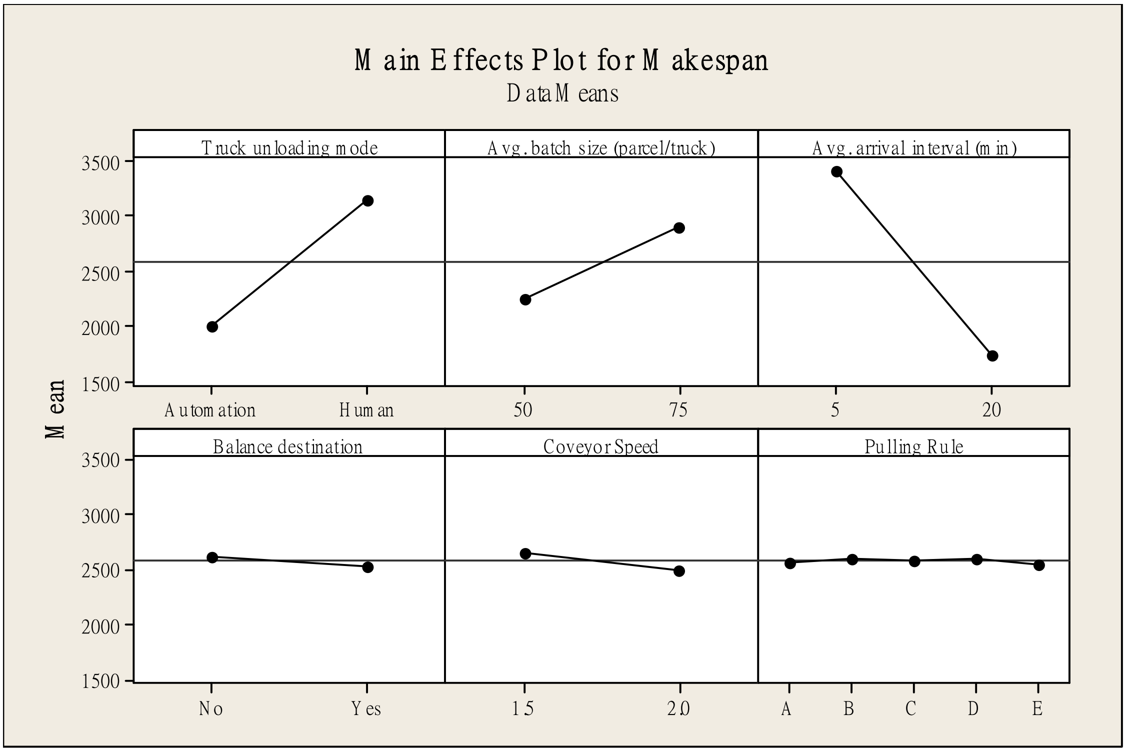


Fig. 16. Main effects plot of DOE.

Table 9
Optimal setting of strategy factors and ASC design factors for minimizing FV

Situation factor		Strategy Factors		ASC Design Factors	
Avg. arrival-time interval (min)	Balance destination	Avg. batch size (parcel/truck)	Truck unloading mode	Conveyor Speed	Pulling Rule
5	Yes	50	Automation	2	A
5	No	50	Automation	2	A
20	Yes	75	Automation	2	C
20	No	75	Automation	2	E

Table 10
Optimal setting of strategy factors and ASC design factors for minimizing FV1-makespan

Situation factor		Strategy Factors		ASC Design Factors	
Avg. arrival-time interval (min)	Balance destination	Avg. batch size (parcel/truck)	Truck unloading mode	Conveyor Speed	Pulling Rule
5	Yes	50	Automation	2	C
5	No	50	Automation	2	A
20	Yes	75	Automation	2	C
20	No	75	Automation	2	A

Table 11
Optimal setting of strategy factors and ASC design factors for minimizing FV2-Average truck queueing time

Situation factor		Strategy Factors		ASC Design Factors	
Avg. arrival-time interval (min)	Balance destination	Avg. batch size (parcel/truck)	Truck unloading mode	Conveyor Speed	Pulling Rule
5	Yes	50	Automation	2	A
5	No	50	Automation	2	A
20	Yes	50	Automation	2	C
20	No	50	Automation	2	A

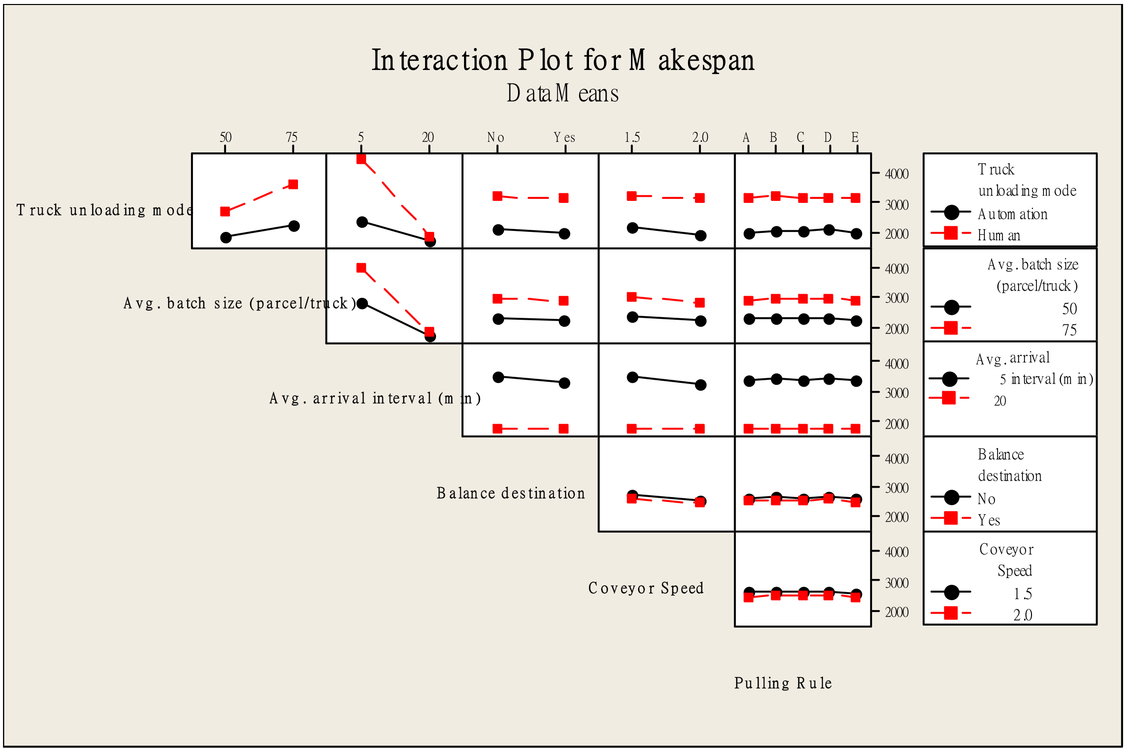


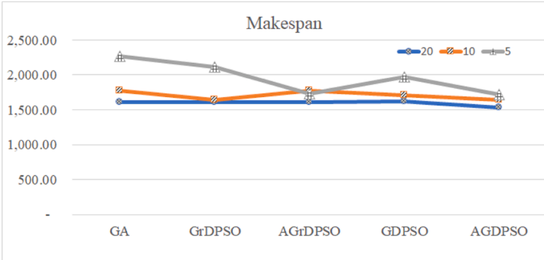
Fig. 17. Interaction plot of DoE.

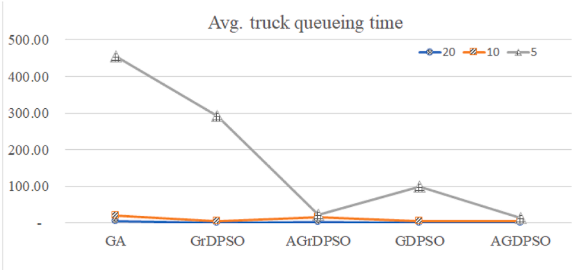
Approach abbreviation	Detail of approach content
GA	Traditional GA with uniform crossover and pairwise interchange mutation.
GrDPSO	Generate velocity array according to Gbest and Pbest and refer to the array for uniform mutation.
AGrDPSO	GrDPSO with time-varying acceleration coefficients
GDPSO	Uniform crossover with Gbest and Pbest and pairwise interchange mutation.
AGDPSO	GDPSO with time-varying acceleration coefficients

A.2. Comparison of processes and mechanisms between algorithms

	Mechanisms				
Process	GA	GrDPSO	AGrDPSO	GDPSO	AGDPSO
Randomly initialization	✓	✓	✓	✓	✓
Evaluation	✓	✓	✓	✓	✓
Encoding	✓	✓	✓	✓	✓
Optimal reference process	-	Mutation	Mutation	Crossover	Crossover
Velocity Array	-	✓	✓	×	×
Crossover	Uniform	Uniform	Uniform	Uniform	Uniform
Mutation	Swap	Uniform	Uniform	Swap	Swap
TVAC	×	×	✓	×	✓

A.3. Performance comparison among different algorithms on makespan and truck queueing time





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